output voltage sequence management, protection circuit fault flag, and switching frequency transition. The fast transient response of this PMIC supports a variable refresh rate. This feature greatly reduces power consumption and enables a smooth and comfortable viewing experience for games and high-quality videos at refresh rates of 60 ~ 240Hz.
Magnachip
https://www.magnachip.com

Industry's first miniature wireless AI vision and audio module

Katana Connect is the industry's first ultra-miniature system on module (SoM) to deliver low-power, AI-enabled computer vision and audio sensing capability with integrated Wi-Fi/Bluetooth connectivity. The SoM is a ready-to-use solution that enables rapid implementation of AI sensing features for connected IoT devices targeting a range of applications, including home automation, security, ambient assisted living, industrial safety, and monitoring. The SoM can also serve as a platform for the development of innovative sensing methods, such as wireless sensing, that complement and enhance Katana's vision and audio edge AI features.

Synaptics Incorporated
https://www.synaptics.com

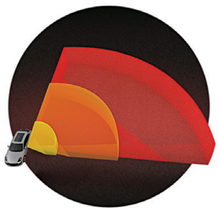
Deterministic processor for ultra-low latency

GroqChip processor accelerates AI, ML, and HPC workloads and removes extraneous circuitry from the chip. The processor reduces data movement for predictable low-latency performance. This standalone chip provides flexible integration into compute-intensive applications. The chip features up to 80TB on-die memory bandwidth that facilitates massive concurrency and data parallelism needed for bandwidth sensitive applications. The chip has integrated software control units at strategic points to optimise data movement and processing. It offers up to 750TOP, 188TFLOP (INT8, FP16 @900MHz). It also offers end-to-end on-chip protection that improves uptime and reliability with error correction code protection throughout the entire GroqChip data path.

Groq
https://groq.com/products

Industry's first software-definable flash lidar

T30P from Preact Technologies is a software-definable flash lidar. It is one of the fastest flash lidars on the market, which makes it suitable for ground and air robotics or industrial applications. It is based on the continuous wave time-of-flight technique, which enables it to work in all weather conditions and detect objects even in the darkest of the night or the most intense direct sunlight. The sensor, when combined with image processing and machine learning, can detect and track multiple objects simultaneously. The sensor has a wide field of view of 120° × 80° and a high operating frequency of 150Hz, which makes it suitable for use in the pre-crash system to deploy airbag even before impact at speeds of over 150kmph.



PreAct Technologies
https://www.preact-tech.com

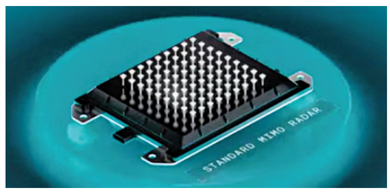
Processor that can provide native dataflow processing

SambaNova DataScale is built with a reconfigurable dataflow unit, the industry's next-generation processor that can provide native dataflow processing. SambaNova offers a new AI system that triples previous system speed. It is a fully integrated hardware-software system that enables organisations to train and deploy demanding deep learning, foundation model, and AI for Science workloads. SambaNova DataScale features SambaFlow, a complete software stack designed to take input from standard machine learning frameworks. It delivers fast performance with even the largest and most challenging models, such as GPT. DataScale reduces the cost and complexity of AI initiatives by providing a fully integrated system, from silicon to software, that is designed to deliver a unique combination of high performance, unprecedented ease of use, and seamless scalability.

SambaNova Systems
https://sambanova.ai

4D imaging radar architecture for autonomous mobility systems

Ambarella's centrally processed 4D imaging radar architecture can enhance safety and driveability of autonomous vehicles. It is the world's first centralised 4D imaging radar architecture, which allows both central processing of raw radar data and deep, low-level fusion with other sensor inputs, including cameras, lidar, and ultrasonics. This architecture will enable higher performance imaging radar systems and new ADAS/AD features while simultaneously optimising the cost of radar sensing. The 4D imaging radar architecture is suitable for use in level 2+ to level 5 autonomous vehicles, as well as autonomous mobile robots (AMRs) and automated guided vehicle (AGV) robots.



Ambarella
https://www.ambarella.com